were the Bell P-39N and P-63A airplanes, the Grumman F6F-3 airplane, the Lockheed P-38F and YP-80A airplanes, and the North American P-51B airplane. The P-38F, P-39N, and the F6F-3 are equipped with NACA conventional airfoils; the P-63A and YP-80A are provided with NACA low-drag airfoils, and the P-51B has a North American Aviation – NACA compromise low-drag airfoil. Two-view drawings of the test aircraft together with pertinent specifications are shown in figure 1. Photographs of the test airplanes are presented in figures 2(a) through 2(f), and wing-root and -tip airfoil sections for these airplanes are presented in figure 3.

The P-51B and the YP-80A airplanes had very carefully filled, waxed, and polished surfaces. The other airplanes were painted with standard camouflage paint. Of the six airplanes tested, the P-38F and P-39N airplanes had the roughest finish and the most openings in the wings.

Standard NACA photographically recording flight instruments were used to measure as a function of time the following variables: indicated airspeed, pressure altitude, normal acceleration, pitching and rolling velocities. The airspeed heads used for all the airplanes tested were of the freely swiveling type, to minimize errors due to the large angles of attack encountered in the tests, and with the exception of the P-38F airplane were mounted on booms extending approximately a chord length or more ahead of the leading edge near the wing tip. The airspeed head of the P-38F airplane was mounted on a boom extending ahead of the fuselage nose. The installations were calibrated for position error.

The free-air temperatures were obtained either from radiosonde observations or from readings of calibrated free-air-temperature indicators or recorders in the airplane.

Photographs of the 0.167-, 0.350-, and 0.333-scale models of the P-38, P-39N, and XP-80 airplanes, respectively, as tested in the Ames 16-foot high-speed wind tunnel are shown in figures 2(g), 2(h), and 2(i). As may be seen in these photographs, the XP-80 model was a complete model; the P-38 model was complete except for the propellers; and the P-39N model was complete except for the propeller and tail. The forces and moments were recorded by self-balancing, recording beam scales.

The airfoil tests conducted in the Ames 1- by 3½-foot high-speed wind tunnel were of 6-inch-chord metal models mounted so as to span completely the 1-foot width of the

tunnel test section. The lift was obtained either from integration of pressure distributions over the airfoils or from measurements of the reactions on the tunnel walls of the forces experienced by the airfoils.

TEST PROCEDURE

All flight data in this report were obtained in gradual stalls made during turns or pull-ups with the airplanes in the clean condition (flaps and landing gear up). With the five propeller-driven airplanes, all stalls were made with the engine throttled and the propeller in the high-pitch setting. Although the power was on during the stalls of the jet-propelled YP-80A airplane, the data were considered similar to the power-off condition of the propeller-driven airplanes as no slipstream existed over the wing.

The flight maximum lift coefficients were calculated by the following formula:

$$C_{L_{max}} = \frac{WA_Z}{qS}$$

where

$C_{L_{max}}$ maximum lift coefficient

W weight of the airplane at the time of stall, pounds

A_Z normal acceleration factor, the ratio of the net aerodynamic force along the airplane Z-axis at the stall (positive when directed upward), to the weight of the airplane

S wing area, including area extending through the fuselage and nacelles, square feet

q dynamic pressure, pounds per square foot

An error of less than two percent was caused by the assumption in the preceding formula that the lift was equal to the normal force WA_Z.

Throughout this report, the Reynolds number was computed

using the mean aerodynamic chord as the characteristic length.

The maximum lift coefficient obtainable in flight may be limited by the effects resulting from a localized stall over a limited portion of the wing or from distortion of the lift curve by compressibility effects; these may be manifested by unstable motions of the airplane, by loss of control effectiveness, or by buffeting of the airplane or controls. Since the extent of such a stall may not be sufficient to prevent attaining higher lift coefficients at greater angles of attack in the wind tunnel, a difference between the maximum lift coefficient obtainable in flight tests and in wind-tunnel tests will usually be anticipated. The characteristics assumed to indicate the stall for each of the six airplanes tested are summarized in the following table:

Airplane	Low Mach numbers	High Mach numbers
F6F-3	Roll-off and slight buffeting	Moderate buffeting with pitch-down, followed by porpoising motions
P-38F	Slight buffeting	Buffeting. Pilot reported apparent ineffectiveness of elevators to increase lift coefficient
P-39N	Buffeting and roll-off	Buffeting and pitch-down
P-51B	Buffeting and roll-off	Severe buffeting with mild roll-off
P-63A	Roll-off and pitch-down	Abrupt roll-off
YP-80A	Roll-off with slight buffeting	Slight buffeting

At low Mach numbers, it is shown that the stalls of all the airplanes except the P-38F were characterized by roll-off. Since stalls characterized by roll-off are clearly defined by the motions of the airplane, the maximum lift coefficients obtained in flight are relatively independent of piloting technique and the amount of control available, and would probably be similar to the values measured in a wind tunnel. At high Mach numbers, however, the stalls of only two airplanes were characterized by roll-off, those of the remaining airplanes

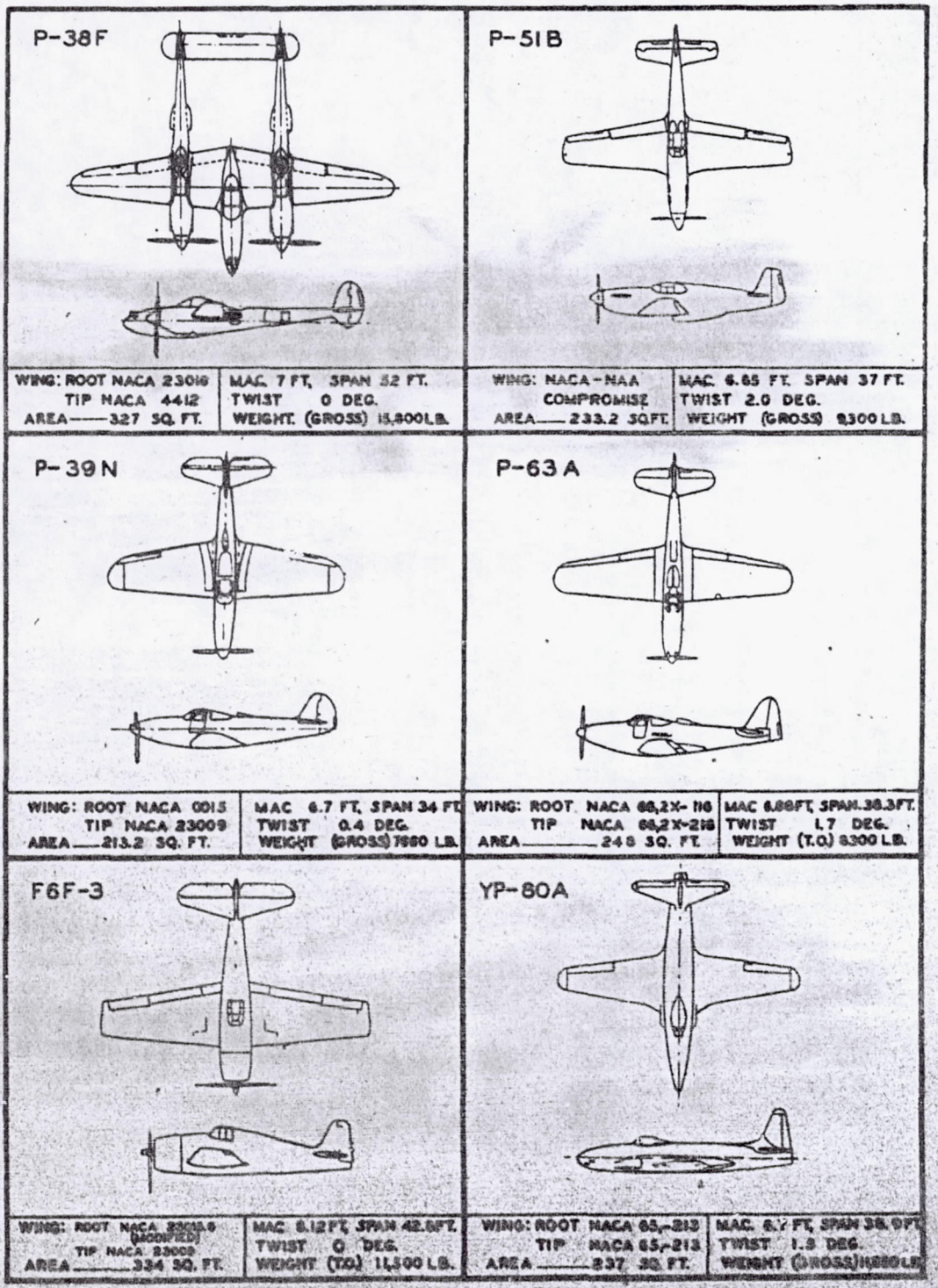


FIGURE 1.- TWO-VIEW DRAWINGS AND PERTINENT SPECIFICATIONS OF THE AIRPLANES TESTED IN FLIGHT.

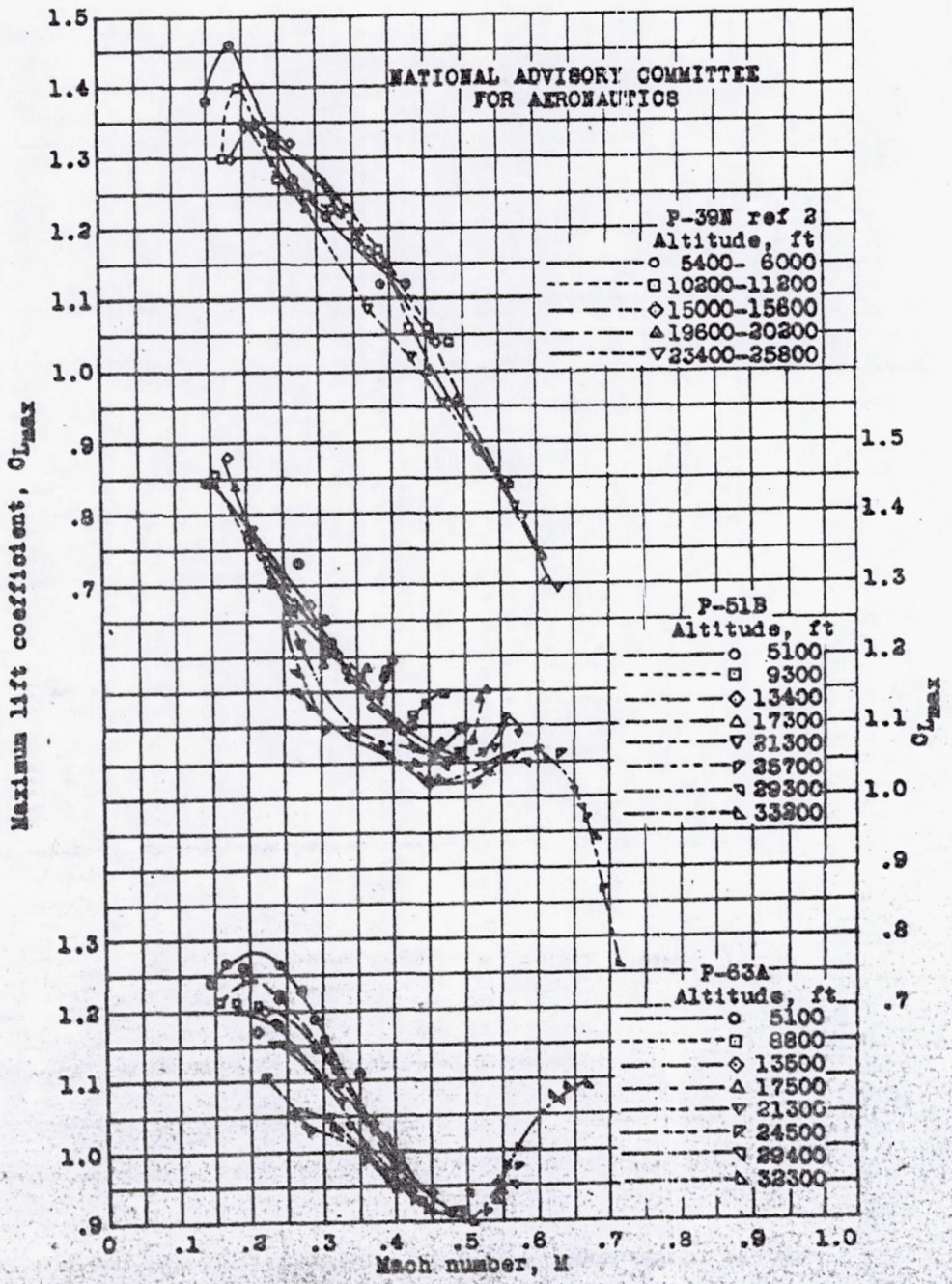


Figure 6.- Variation of maximum lift coefficient with Mach number for various altitudes.